

D = event of having disease
 D^c = event of not having disease
 A = event that test A is positive
 B = event that test B is positive

$$\begin{aligned}
 a. P(D|A \cap B) &= \frac{P(A \cap B|D)P(D)}{P(A \cap B|D)P(D) + P(A \cap B|D^c)P(D^c)} = \frac{P(A|D)P(B|D)P(D)}{P(A|D)P(B|D)P(D) + P(A|D^c)P(B|D^c)P(D^c)} \\
 &= \frac{(.9)(.95)(.01)}{(.9)(.95)(.01) + (.06)(.08)(.99)} = \frac{.00855}{.013302} = \boxed{.64276}
 \end{aligned}$$

$$\begin{aligned}
 b. P(B|A) &= \frac{P(B \cap A)}{P(A)} = \frac{P(B \cap A|D)P(D) + P(B \cap A|D^c)P(D^c)}{P(A|D)P(D) + P(A|D^c)P(D^c)} \\
 &= \frac{(.9)(.95)(.01) + (.06)(.08)(.99)}{(.9)(.01) + (.06)(.99)} = \frac{.013302}{.0684} \\
 &= \boxed{.19447}
 \end{aligned}$$

$$\begin{aligned}
 c. P(A \cup B|D) &= P(A|D) + P(B|D) - P(A \cap B|D) \\
 &= P(A|D) + P(B|D) - P(A|D)P(B|D) \\
 &= (.9) + (.95) - (.9)(.95) \\
 &= \boxed{.995}
 \end{aligned}$$

$$1.38 \text{ (a.) } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{1} = P(A \cap B) = P(A)$$

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c) = P(A|B)P(B) = P(A|B) = P(A \cap B)$$

(b.) $A \subset B$ implies $A \cap B = A$.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$$

$$1.38 \text{ (c.) } P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)}$$

mutually
exclusive

$$(d.) P(A \cap B \cap C) = P[A \cap (B \cap C)] = P(A|B \cap C)P(B \cap C)$$

$$= P(A|B \cap C)P(B|C)P(C)$$

by conditional probability formula.

1.39 (a.) If A and B are mutually exclusive then $P(A \cap B) = 0$.

In order for A and B to be independent, we must have

$P(A \cap B) = P(A)P(B)$ but this is impossible since $P(A) > 0$ and $P(B) > 0$

(b.) If A and B are independent then $P(A \cap B) = P(A)P(B) > 0$

(since $P(A) > 0$ and $P(B) > 0$). This means that $P(A \cap B) \neq 0$

so A and B cannot be mutually exclusive.

1.40 a. $P(A^c \cap B) = P(A^c|B)P(B) = [1 - P(A|B)]P(B) = [1 - P(A)]P(B) = P(A^c)P(B)$, where the third equality follows from the independence of A and B .

b. $P(A^c \cap B^c) = P(A^c) - P(A^c \cap B) = P(A^c) - P(A^c)P(B) = P(A^c)P(B^c)$.

$$\begin{aligned}
 5. \quad A, B, C \text{ mutually independent} &\Rightarrow P(A \cap B \cap C) = P(A)P(B)P(C) \\
 &P(A \cap B) = P(A)P(B) \\
 &P(A \cap C) = P(A)P(C) \\
 &P(B \cap C) = P(B)P(C)
 \end{aligned}$$

$$(i) \quad A \cap B \text{ and } C \text{ are independent if } P((A \cap B) \cap C) = P(A \cap B)P(C)$$

$$\begin{aligned}
 P((A \cap B) \cap C) &= P(A \cap B \cap C) = P(A)P(B)P(C) = P(A \cap B)P(C) \quad \square \\
 &\quad \uparrow \qquad \qquad \qquad \uparrow \\
 &\quad \text{by mutual independence}
 \end{aligned}$$

$$(ii) \quad A \cup B \text{ and } C \text{ are independent if } P((A \cup B) \cap C) = P(A \cup B)P(C)$$

$$\begin{aligned}
 P((A \cup B) \cap C) &= P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - P((A \cap C) \cap (B \cap C)) = P(A \cap C) + P(B \cap C) + P(A \cap B \cap C) \\
 &= P(A)P(C) + P(B)P(C) + P(A)P(B)P(C) = (P(A) + P(B) - P(A)P(B))P(C) \\
 &\quad \uparrow \\
 &\quad \text{by mutual independence} \\
 &\quad \downarrow \\
 &= (P(A) + P(B) - P(A \cap B))P(C) = P(A \cup B)P(C) \quad \square
 \end{aligned}$$

1.47 All of the functions are continuous, hence right-continuous. Thus we only need to check the limit, and that they are nondecreasing

- a. $\lim_{x \rightarrow -\infty} \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x) = \frac{1}{2} + \frac{1}{\pi} \left(-\frac{\pi}{2}\right) = 0$, $\lim_{x \rightarrow \infty} \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x) = \frac{1}{2} + \frac{1}{\pi} \left(\frac{\pi}{2}\right) = 1$, and $\frac{d}{dx} \left(\frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)\right) = \frac{1}{1+x^2} > 0$, so $F(x)$ is increasing.
- b. See Example 1.5.5.
- c. $\lim_{x \rightarrow -\infty} e^{-e^{-x}} = 0$, $\lim_{x \rightarrow \infty} e^{-e^{-x}} = 1$, $\frac{d}{dx} e^{-e^{-x}} = e^{-x} e^{-e^{-x}} > 0$.
- d. $\lim_{x \rightarrow -\infty} (1 - e^{-x}) = 0$, $\lim_{x \rightarrow \infty} (1 - e^{-x}) = 1$, $\frac{d}{dx} (1 - e^{-x}) = e^{-x} > 0$.
- e. $\lim_{y \rightarrow -\infty} \frac{1-\epsilon}{1+e^{-y}} = 0$, $\lim_{y \rightarrow \infty} \epsilon + \frac{1-\epsilon}{1+e^{-y}} = 1$, $\frac{d}{dy} \left(\frac{1-\epsilon}{1+e^{-y}}\right) = \frac{(1-\epsilon)e^{-y}}{(1+e^{-y})^2} > 0$ and $\frac{d}{dy} \left(\epsilon + \frac{1-\epsilon}{1+e^{-y}}\right) > 0$, $F_Y(y)$ is continuous except on $y = 0$ where $\lim_{y \downarrow 0} \left(\epsilon + \frac{1-\epsilon}{1+e^{-y}}\right) = F(0)$. Thus is $F_Y(y)$ right continuous.

1.53 a. $\lim_{y \rightarrow -\infty} F_Y(y) = \lim_{y \rightarrow -\infty} 0 = 0$ and $\lim_{y \rightarrow \infty} F_Y(y) = \lim_{y \rightarrow \infty} 1 - \frac{1}{y^2} = 1$. For $y \leq 1$, $F_Y(y) = 0$ is constant. For $y > 1$, $\frac{d}{dy} F_Y(y) = 2/y^3 > 0$, so F_Y is increasing. Thus for all y , F_Y is nondecreasing. Therefore F_Y is a cdf.

b. The pdf is $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} 2/y^3 & \text{if } y > 1 \\ 0 & \text{if } y \leq 1. \end{cases}$

c. $F_Z(z) = P(Z \leq z) = P(10(Y - 1) \leq z) = P(Y \leq (z/10) + 1) = F_Y((z/10) + 1)$. Thus,

$$F_Z(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ 1 - \left(\frac{1}{[(z/10)+1]^2}\right) & \text{if } z > 0. \end{cases}$$

X 8.

~~10.~~

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{x^2}{2} & 0 < x < 1 \\ 2x - \frac{x^2}{2} - 1 & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

Note that

- (a) $\lim_{x \rightarrow -\infty} F(x) = 0$ since $F(x) = 0$ for all $x \leq 0$, Similarly, $\lim_{x \rightarrow \infty} F(x) = 1$ since $F(x) = 1$ for all $x \geq 2$,
- (b) Note that the pdf $f(x) = \frac{dF(x)}{dx} \geq 0$. Thus, the CDF is nondecreasing.
- (c) Not only is $F(x)$ right continuous, it is continuous for all x . We show right continuity from the following:

1. $\lim_{x \downarrow 0} F(x) = \lim_{x \downarrow 0} \frac{x^2}{2} = 0$
2. $\lim_{x \downarrow 1} F(x) = \lim_{x \downarrow 1} 2x - \frac{x^2}{2} - 1 = 1/2$.
3. $\lim_{x \downarrow 2} F(x) = 1$ since $F(x) = 1$ for all $x \geq 2$.
4. The function is continuous in all other areas because they are simply polynomials.

9.

$$f_X(x) = \begin{cases} \frac{x}{4} & 0 < x \leq 2, \\ e^{-2(x-2)} & x > 2 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the CDF of X and prove that it is valid.

$$F_X(x) = \begin{cases} \frac{1}{8}x^2 & 0 < x \leq 2 \\ 1 - \frac{1}{2}e^{-2(x-2)} & x > 2 \\ 0 & x \leq 0 \end{cases}$$

$$(a) \lim_{x \rightarrow \infty} F_X(x) = 1 - \frac{1}{2} \lim_{x \rightarrow \infty} e^{-2(x-2)} = 1$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$(b) \text{ For } x' > x, F_X(x') - F_X(x) = \begin{cases} \frac{1}{8}x' - \frac{1}{8}x > 0 & 0 < x \leq 2 \\ e^{-2(x'-2)} - e^{-2(x-2)} > 0 & \text{for } x > 2 \end{cases}$$

non-negative

$$(c) \lim_{x \downarrow x_0} F_X(x) = F_X(x_0) \text{ because } F_X(x) \text{ is continuous everywhere}$$

(b) Is X a continuous random variable? Why or why not?

Yes, CDF is continuous everywhere.

10.

So, they draw this picture. This random variable is neither discrete, nor continuous. It is continuous from $(0,1)$, but there is a point mass at 0 that has probability $\frac{1}{2}$. Basically, the probability of getting 0 is $\frac{1}{2}$, while the rest of the random variable behaves continuously between $(0,1)$.

11.

$$a. f(x) = P(X=x) = \frac{c}{2^x} \text{ for } x=0,1,2,\dots$$

$$1 = \sum_{x=0}^{\infty} \frac{c}{2^x} = c \sum_{x=0}^{\infty} \left(\frac{1}{2}\right)^x = c \left(\frac{1}{1-\frac{1}{2}}\right) = c \cdot 2$$

geometric series

$$\boxed{c=2}$$

$$b. \int_0^1 cx^2 dx = 1$$

$$c \cdot \left(\frac{1}{3}x^3\right) \Big|_0^1 = 1$$

$$\frac{c}{3} = 1$$

$$\boxed{c=3}$$

12.

$$(a) \int_0^1 c x^2 dx = 1$$

$$c \left[\frac{1}{3} x^3 \right]_0^1 = 1$$

$$c/3 = 1$$

$$\boxed{c = 3}$$

$$(b) F_X(x) = \int_0^x 3t^2 dt = t^3 \Big|_0^x = x^3$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^3 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

$$(c) P(0.1 \leq X \leq 0.5) = F_X(0.5) - F_X(0.1) = 0.5^3 - 0.1^3 = \boxed{0.124}$$

13. CDF of exponential (rate parameter λ). $F_X(x) = 1 - e^{-\lambda x}$

$$(a) P(X < 10) = F_X(10) = 1 - e^{-0.1(10)} = \boxed{1 - e^{-1} = 0.632121}$$

$$(b) P(5 < X < 15) = F_X(15) - F_X(5) = (1 - e^{-0.1(15)}) - (1 - e^{-0.1(5)})$$

$$= \boxed{e^{-0.5} - e^{-1.5} = 0.3834}$$

$$(c) P(X > t) = 0.01 \Rightarrow 1 - F_X(t) = 0.01 \Rightarrow e^{-0.1t} = 0.01 \Rightarrow \boxed{t = \frac{-\ln 0.01}{0.1} = 46.0517}$$

$$14. F_X(1) = 0.05 \Rightarrow 1 - e^{-\lambda} = 0.05 \Rightarrow \boxed{\lambda = -\ln 0.95 = 0.0512933}$$